

Recommendation System for Elective Courses using Content-based Filtering and Weighted Cosine Similarity

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Abstract—Each study program requires students to take several elective courses. The appropriateness of the elective courses taken with the student's abilities can be one of the factors for the success of student studies. This research focuses on building a content-based filtering recommendation system that provides several elective courses recommendation according to the student's academic history. The proposed recommendation systems' results are based on preprocessed word items from courses taken by the user. The weighted cosine similarity between the elective courses syllabus and the user profiles is calculated. Moreover, the experiment employed a dataset of the CSUGM course syllabus. The proposed recommendation system is evaluated in two ways, i.e., questionnaire method and validation method. The questionnaire method obtains an assessment of system performance, hence the validation method to get the average accuracy. The questionnaire was conducted by involving thirty students of the CSUGM undergraduate program. The experimental results show that the proposed recommendation system has a good performance proven by the percentage of recommendation diversity 81.67%. Furthermore, the accuracy of the proposed recommendation system has an average of 64%.

Keywords—*recommendation system, elective courses, content-based filtering, cosine similarity*

I. INTRODUCTION

A course is a unit of lessons taught at the tertiary level and similar to a subject at primary and secondary school levels. Every student requires to fulfill certain credits of a set of courses that consist of compulsory and elective courses. However, some students have difficulty choosing elective courses according to their interests and abilities [1]. Even though the portion of elective courses is smaller than compulsory ones, they have a considerable role in determining the academic achievement index of each student [2]. Thus, choosing appropriate elective courses is important to all student success in carrying out the academic process.

The recommendation system [3] can be a solution in helping students determine the elective courses they want to choose. A good design in making this recommendation system is to consider particular characteristics of students so that it can produce recommendations for elective courses that are following student interests and abilities [4]. A recommendation system requires to provide alternative considerations that are useful to support student decisions in determining the elective courses to be taken [5].

Previous researchers have proposed and developed an elective courses recommendation system, such as [4], [5], and [6]. However, no research considers the historical attribute of course grades from students on their proposed elective courses recommendation system.

Therefore, this research focuses on building a recommendation system for elective courses based on the history of student study results. The history of study results used is not only the name of the course but also the grade. The elective courses come from the curriculum of the study program minus all taken elective courses by the student. Next, all the taken courses by students are utilized to retrieve the elective courses. The peculiarity of the research is reflecting the student's score as the weight of the courses.

Furthermore, this research evaluated the performance of the proposed recommendation system objectively and subjectively. The accurate level is used to evaluate the system objectively. Whereas relevancy, novelty, serendipity, and increasing recommendation diversity rate are utilized to analyze the quality of the proposed system subjectively [3]. The relevance measures the delivery of the proposed system whether appropriate or inappropriate to the user requirements. The novelty evaluation is to measure whether the system provides new elective courses that the user has never predicted. The serendipity evaluates whether the system delivers unexpected but interesting elective courses. An increasing recommendations diversity rate measures whether the system is capable of providing diverse elective courses. Moreover, this study employed thirty students as response to evaluate the proposed system.

In section 2, some methodologies such as text preprocessing and feature approaches are described. Furthermore, data, schema, and results of the experiment are presented in section 3. Some conclusions and future works are given in section 4.

II. METHODS

A. Recommendation System

This research focuses to proposed the content-based filtering elective courses recommendation system. The Content-based filtering method relies heavily on individual/user attributes to provide output recommendations. Furthermore, the content-based filtering method is the best method to build a system with a text dataset [7]. The proposed content-based filtering recommendation system focuses to

provide a list of elective courses to students based on their academic histories.

Figure 1 presents the proposed recommendation system. It requires two data sources, namely the curriculum, and student transcripts. We only consider the course names and courses topics, which are called a syllabus.

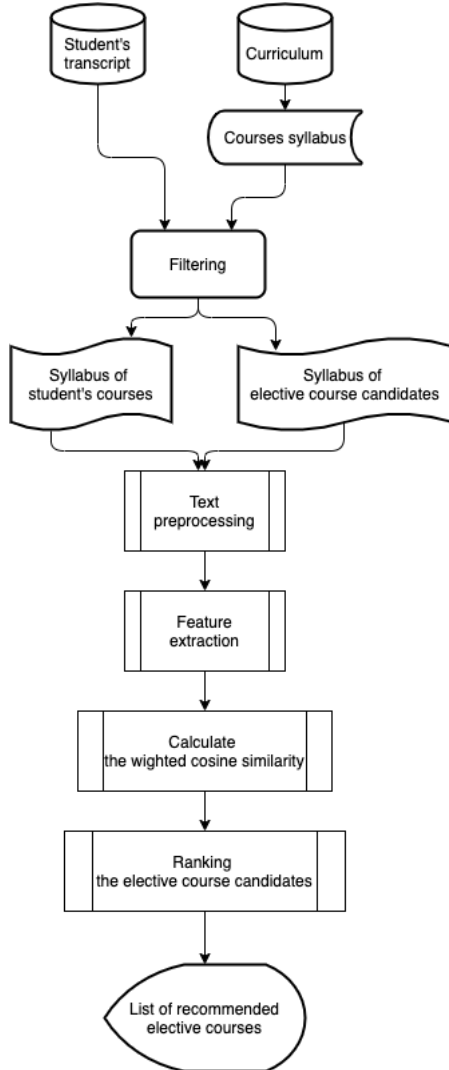


Figure 1 Architecture of Elective Courses Recommendation System

Furthermore, the syllabus is divided into two, namely the syllabus of taken courses, both compulsory and elective, and the syllabus of the elective course candidates. Since the two sets of the syllabus are text data, so we did text preprocessing. Furthermore, TF-IDF is performed to extract the result of text preprocessing. The process of determining candidates for elective courses that are similar to student profiles uses a weighted cosine similarity calculation. The five candidates for elective courses with the most similarity are the results of the recommendations. Thus, the list of recommended elective courses is obtained by calculating the weighted cosine similarity between the curriculum profile and the student profile.

B. Text Preprocessing

Text Preprocessing is a series of cleaning and preparation processes before text/documents are processed or used for further processing. Text preprocessing is the most important part of Natural Language Processing [8]. With this text

processing, it is hoped that it can get rid of unnecessary words and maximize the data processing that will be carried out. The purpose of doing text preprocessing is also to improve classification performance and speed up time in the data processing.

The proposed recommendation system performs text preprocessing (Fig 2) on both syllabus courses, syllabus of student's courses and syllabus of elective course candidates, i.e., case folding, punctuation removal, tokenization, and stop words removal.

A text document has multiple characters (both letters, punctuation, and numbers). So that in this research, the case-folding process aims to change all letters in a text document to lowercase. Hence the punctuation removal deletes any characters other than letters as well as a delimiter. Next, the tokenization in the part of the text preprocessing is a procedure to split a text into words, phrases, or other meaningful parts, namely tokens. In other words, tokenization is a form of text segmentation [9].

Stopwords are words that are not descriptive and meaningless with a high rate of occurrence in a dataset. These words don't need to be analyzed because they don't contain useful information. Stopword removal is the stage of deleting these stopwords so that the remaining words are important words that have meaning.

Next, lemmatization is a process in text preprocessing that determines the form of a word into a root word or looks for the root word of each based on the context of the sentence [10]. The purpose of the lemmatization process is to optimize the text mining process. The lemmatization process has the same goal as the stemming process, which is the same as finding the root of a word. The process followed by performing feature extraction of the text preprocessing results using the TF-IDF method.

C. Term Frequency – Inverse Document Frequency

The term is a collection of unique words of preprocessing results on the syllabus. Furthermore, *Term Frequency* tf is defined (Eq. 1) as the number of a unique word occurrences in a syllabus document. Thus, $tf(w_i, d_j)$ is determined as the frequency of the word w_i in the syllabus document d_j

$$tf(w_i, d_j) = f(w_i, d_j) \quad (1)$$

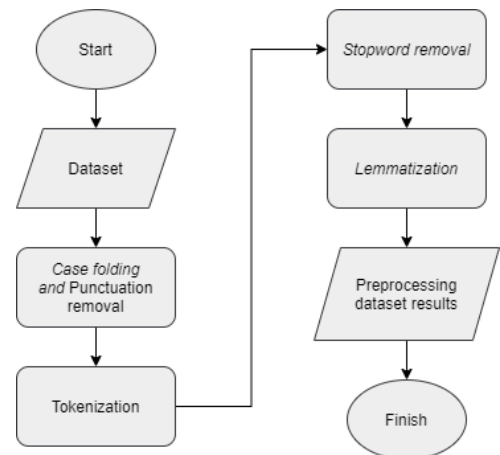


Figure 2 Text Preprocessing Flow

Assume that s is the number of syllabus documents and td_j is the collection of terms in syllabus document j . and $j = 1, 2, \dots, s$, while the number of terms, n refers to total terms uniquely from s documents, namely $n = |td_1 \cap td_2 \cap \dots \cap td_s|$.

Next, inverse document frequency (idf) is the inverse frequency of documents containing the term. IDF is a reduction of the term dominance, so the smaller number of documents containing the term, the larger the IDF value of the term. IDF is calculated using Eq. 2, where s is the total number of documents and $s(w_i)$ is total documents containing the unique term i .

$$idf(w_i) = \log\left(\frac{s}{s(w_i)}\right) \quad (2)$$

Last, the feature of the documents, i.e., term frequency-inverse document frequency ($tfidf$) is extracted. The $tfidf$ is a numerical statistic that reflects how important a word is to a document. The $tfidf$ value increases proportionally to the number of times a word appears in the document [12]. The formula used to calculate $tfidf$ of term i on document j is in Eq. 3, where tf_{ij} is the term frequency of term i in document j and $idf(w_i)$ is the inverse document frequency of term i .

$$tfidf(w_i, d_j) = tf(w_i, d_j) \times idf(w_i) \quad (3)$$

$$tfidf(w_i, d_j) = tf(w_i, d_j) \times \left(\log\left(\frac{s}{s(w_i)}\right) + 1\right)$$

D. Weighting Cosine Similarity

The $tfidf$ value is a representation of the feature extraction results of the syllabus document. Cosine similarity measurement is utilized to calculate the similarity between the syllabus of student's courses and the syllabus of elective course candidates.

The similarity between syllabus of student's courses k , q_k and the syllabus of elective course candidates j , p_j is defined in Eq. 4. Next, the results of the similarity calculation will be used to sort all the matched videos from highest to lowest and taken the top five videos as the result of the proposed system.

$$Sim(q_k, p_j) = \frac{\sum_{i=1}^n tfidf(w_i, q_k) \times tfidf(w_i, p_j)}{\sqrt{\sum_{i=1}^n (tfidf(w_i, q_k))^2} \times \sqrt{\sum_{i=1}^n (tfidf(w_i, p_j))^2}} \quad (4)$$

Because this study proposes considering the student's score, we adjust the cosine similarity equation (Eq. 4) into weighted cosine similarity. The weight is the result of normalization from the conversion of the alphabetical value obtained by students to numerical values on a scale of 1 to 4, $Score(q_k)$. Thus, the weighted cosine similarity between student courses and candidate elective courses, $WSim(q_k, p_j)$ is defined as Eq. 5.

$$WSim(q_k, p_j) = Sim(q_k, p_j) \times \frac{Score(q_k)}{4} \quad (5)$$

E. Ranking

The list of recommended elective courses is based on the value of the correlation of the candidate elective courses with the student's abilities. The correlation value, $Cor(p_j)$ is defined in Eq. 6 as the average similarity between each elective course candidate and each student course, where m is the number of student's courses.

$$Cor(p_j) = \frac{\sum_{k=1}^m WSim(q_k, p_j)}{m} \quad (6)$$

Then all candidates for elective courses are ranked in descending order based on their correlation value. A higher correlation value represents the smallest rank. The list of recommended elective courses contains elective course candidates ranked 1 to 7.

III. RESULTS AND DISCUSSION

A. Dataset

The dataset used in this research is subject data such as the name of the course and the syllabus of the 2016 curriculum for the UGM Computer Science study program. Course data and syllabus are stored hardcoded into tables as shown in Fig. 3.

The stored course data is then preprocessed to produce a lean syllabus. The results of the preprocessing syllabus are then stored in a new column in the previous table as in Fig. 4.

B. Text Processing on Grade Transcript Files

Before the calculation process begins, the system first accepts input in the form of a student transcript file. In the file, there are attributes that are used in the calculation process such as course code, course name, and the value that has been obtained. After the system accepts input, the system will perform text processing such as changing the course value which was originally a letter-scale grade to a number-scale value 4 and also filtering out any elective courses that the user has not taken. Fig. 5 represents an example of the results of text processing in the transcript file that has been entered into the system.

A	B	C	D
No	Kode MK	Nama MK	Silabus
1	MFF1011	Basic Physics I	Measurement and Magnitude of Physics Kinematics Dynamics I: The Concept of Style Dynamics II: B
2	MII1003	Elementary Linear Algebra	Systems of linear equations and solutions, Gauss-Jordan Elimination (Elementary Line Operations)
3	MII1201	Programming I	Programming 1 course provides students with knowledge and skills to analyze problems, design alg
4	MII1203	Logic for Computer Science	This lecture discusses some basic concepts of logic. The topics provided include propositional logic
5	MII1205	Introduction to Computer Scie	This course provides an introduction to computer science as from technology and science. Topics
6	MII1211	Programming II	Programming II Course is a continuation of the course of Programming I. This course provides kno
7	MII1213	Discrete Mathematics	In this course, some basic concepts of discrete mathematics required in the field of computer scie
8	MII1811	Digital Systems	This lecture presents a discussion of how logical circuits are used to build computers. Subject: Book
9	MII2201	Analysis of Algorithm and Com	In the course of Algorithm and Complexity Analysis, students will be introduced to algorithm theor
10	MII2203	Probability and Stochastic Pro	This lecture provides theoretical and applicative foundations for probability and stochastic proces
11	MII2205	Language and Automaton	In this course, the basic ideas of computer science theory were introduced to students. The subject
12	MII2207	Computer Graphics	This course provides an introduction to computer graphics algorithms, starting from graphic pipeli
13	MII2211	Advanced Algorithm	Algorithm Advanced Algorithm Course is a continuation of the course of Algorithm Analysis and Cc
14	MII2213	Numerical Method	In this course, various methods for numerical solutions of various mathematical problems are intr
15	MII2411	Artificial Intelligence	The main objectives and targets of this lecture are to make students have or retain the basic techn
16	MII2501	Database	Database is a collection of data in the form of information. The purpose of this lecture is to provi
17	MII2601	Microprocessor	This course focuses on the principles and practice of modern embedded systems design. The disc
18	MII2602	Computer Network	Course Computer network is a basic course in the field of digital data communication, especially d
19	MII2603	Organization and Computer Ar	Organizational study and computer architecture studied the development of organization and con
20	MII2611	Operating System	This course is the introduction course of the Operating System (OS). Explain how the operating sys
21	MII3003	Research Method	This course provides an introduction to conceptual and practical research methods for computer i
22	MII3007	Philosophy of Computer Scien	Computer Science Philosophy course is basically answer the question of whether the computer ca
23	MII3411	Machine Learning	This course provides an introduction to the concepts, techniques, algorithms in the learning machi
24	MII3501	Software Development	The software is a business product that requires an engineering approach to make it as from the r
25	MII3601	Cryptography and Network Sec	Cryptography provides an introduction to the basic principles of cryptography, especially digital cri

Figure 2 Pieces of Course Data Collection Results

	A	B	C	D	E	F	G
	No	Kode MK	Nama MK	Silabus	clean_silabus_all		
1	1	MFF1011	Basic Phys	Measurement and measurement magnitude physic kinematics dyn			
2	2	MII1003	Elementar	Systems of linear (system linear equation solution gaussjordan elir			
3	3	MII1201	Programm	Programming 1 co programming course provides student knowledg			
4	4	MII1203	Logic for C	This lecture discuss lecture discuss basic concept logic topic provide			
5	5	MII1205	Introducti	This course provid course provides introduction computer science			
6	6	MII1211	Programm	Programming II Cc programming ii course continuation course prog			
7	7	MII1213	Discrete M	In this course, son course basic concept discrete mathematics requ			
8	8	MII1811	Digital Sys	This lecture prese lecture present discussion logical circuit used bu			
9	9	MII2201	Analysis of	In the course of Al course algorithm complexity analysis student in			
10	10	MII2203	Probability	This lecture provide lecture provides theoretical applicative foundat			
11	11	MII2205	Language	In this course, the course basic idea computer science theory intrc			
12	12	MII2207	Computer	This course provid course provides introduction computer graphic			
13	13	MII2211	Advanced	Algorithm Advanci algorithm advanced algorithm course continuat			
14	14	MII2213	Numerical	In this course, vari course various method numerical solution vario			
15	15	MII2411	Artificial	In the main objectiv main objective target lecture make student reta			
16	16	MII2501	Database	Database is a coll database collection data form information purp			
17	17	MII2601	Microproc	This course focuse course focus principle practice modern embedd			
18	18	MII2602	Computer	Course Computer course computer network basic course field digi			
19	19	MII2603	Organizati	Organizational stu organizational study computer architecture stud			
20	20	MII2611	Operating	This course is the course introduction course operating system o			
21	21	MII3003	Research	This course provide course provides introduction conceptual practic			
22	22	MII3007	Philosophy	Computer Science computer science philosophy course basically an			
23	23	MII3411	Machine	L This course provide course provides introduction concept techniq			
24	24	MII3501	Software	The software is a software business product requires engineering			
25	25	MII3601	Cryptograp	Cryptography provi cryptography provides introduction basic princ			

Figure 3 Cleansed Pieces of Cleared Course Data Results

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[[ 'MMM1102', 'KALKULUS II\rKelas: KAL 3', 2.5],
[ 'MFF1012', 'FISIKA DASAR II\rKelas: FIS-7-MFF1012', 2.25],
[ 'MFF1011', 'FISIKA DASAR I\rKelas: FD-2-MFF1011', 3.0],
[ 'UNU1000', 'AGAMA ISLAM\rKelas: AI-2-UNU1000', 4.0],
[ 'MKK1101', 'KIMIA DASAR I\rKelas: KD-2-MKK1101', 3.0],
[ 'MMM1101', 'KALKULUS I\rKelas: MMM1101-2', 3.25],
[ 'MII1211', 'PEMROGRAMAN II\rKelas: ILKOM-B-MII1211', 3.25],
[ 'MII2501', 'BASIS DATA\rKelas: ILKOM-B-MII2501', 2.75],
[ 'UNU1010', 'PANCASILA\rKelas: ILKOM-B-UNU1010', 4.0],
[ 'MII1203', 'LOGIKA INFORMATIKA\rKelas: ILKOM-B-MII1203', 4.0],
[ 'MII2201', 'ANALISIS ALGORITMA DAN KOMPLEKSITAS\rKelas: ILKOM-B-MII2201', 3.25],
[ 'MII2205', 'BAHASA DAN OTOMATA\rKelas: ILKOM-B-MII2205', 3.25],
[ 'MII2601', 'MIKROPROSESOR\rKelas: ILKOM-B-MII2601', 4.0],
[ 'MII1213', 'MATEMATIKA DISKRIT\rKelas: ILKOM-B-MII1213', 3.5],
[ 'MII1003', 'ALJABAR LINEAR FUNDAMENTAL\rKelas: ILKOM-B-MII1003', 3.25],
[ 'MII1201', 'PEMROGRAMAN I\rKelas: PK-2-MII1201', 3.25],
[ 'MII1202', 'PRAK. PEMROGRAMAN I\rKelas: ILKOM-A1-MII1202', 3.75],
[ 'MII1212', 'PRAK. PEMROGRAMAN II\rKelas: ILKOM-A1-MII1212', 3.75],
[ 'MII1205', 'PENGANTAR ILMU KOMPUTER\rKelas: ILKOM-B-MII1205', 3.25],
[ 'MII2502', 'PRAK. BASIS DATA\rKelas: ILKOM-A2-MII2502', 3.75],
[ 'MII2203', 'PROBABILITAS DAN PROSES STOKASTIK\rKelas: ILKOM-B-MII2203', 3.75],
[ 'MII2207', 'GRAFIKA KOMPUTER\rKelas: ILKOM-B-MII2207', 2.75],
[ 'MII4505', 'PEMROGRAMAN WEB\rKelas: ILKOM-MII4505', 3.0],
[ 'MII2603', 'ORGANISASI DAN ARSITEKTUR KOMPUTER\rKelas: ILKOM-B-MII2603', 3.5],
[ 'MII1811', 'SISTEM DIGITAL\rKelas: ILKOM-B-MII1811', 3.25],
[ 'MII1001', 'BAHASA INGGRIS\rKelas: ILKOM-B-MII1001', 4.0],

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Figure 4 Examples of Text Preprocessing Results on Input Transcripts

C. Recommendation Results

After the system receives the user value transcript data, then the TF-IDF calculation process begins, followed by calculating the cosine similarity between the syllabus of courses that the user has taken and the syllabus of the elective courses that the user has not taken. Then the system displays 2 types of recommendation results, the first is the recommendation result without a weighted value, and the second is the recommendation result with a weighted value. An example of the recommendation results is presented in Fig. 6 and Fig. 7.

D. System Testing Results

System testing was carried out by conducting a survey of 30 students of UGM Computer Science class 2018 and 2019 who were randomly selected. Before filling out the assessment form, students are asked to first try the recommendation system by uploading a temporary transcript file into the system.

In the assessment form that has been provided, there are 2 assessment sections, the first is a section for assessing the results of recommendations for elective courses without weight, and the second section for assessing the results of recommendations for elective courses by weight. In each section, there are 4 questions to assess each attribute of the system test, namely relevance, novelty, serendipity, and increasing recommendation diversity.

1. Capita Selecta on Object Oriented Software Development - Average similarity : 0.00765	▼
2. Network and Mobile Device Programming - Average similarity : 0.00663	▼
3. Capita Selecta on Computational Logics - Average similarity : 0.00659	▼
4. Capita Selecta on Computer and Network Systems - Average similarity : 0.00628	▼
5. Network Analysis and Optimization - Average similarity : 0.00616	▼
6. Data Mining and Business Intelligence - Average similarity : 0.00575	▼
7. Distributed Algorithm - Average similarity : 0.00547	▼

Figure 5 Example of Recommendation Results with Weighted Cosine Similarity

1. Capita Selecta on Object Oriented Software Development - Average similarity : 0.00877	▼
2. Capita Selecta on Computational Logics - Average similarity : 0.00867	▼
3. Network and Mobile Device Programming - Average similarity : 0.00788	▼
4. Capita Selecta on Computer and Network Systems - Average similarity : 0.00722	▼
5. Network Analysis and Optimization - Average similarity : 0.00717	▼
6. Data Mining and Business Intelligence - Average similarity : 0.00674	▼
7. Parallel Programming and Systems - Average similarity : 0.00648	▼

Figure 6 Example of Recommendation Results without Weighted Cosine Similarity

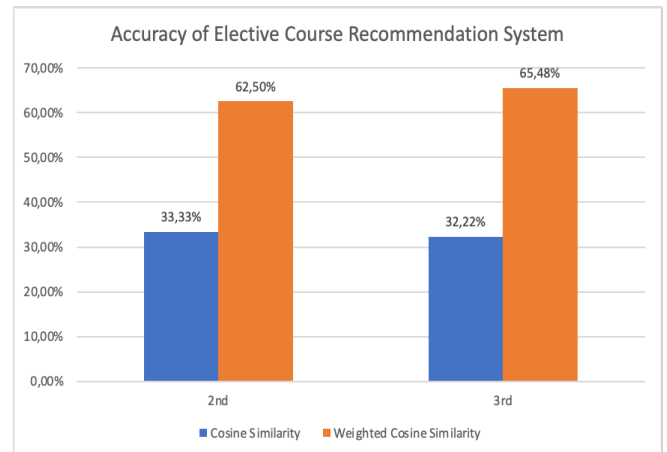


Figure 8 Example of Recommendation Results without Weighted Cosine Similarity

From the system testing, the average results of non-weighted recommendation testing for the relevance attribute were 68%, the novelty was 90.67%, serendipity was 72.67%, and increasing recommendation diversity was 80%. Then for the average results of weighted recommendation testing for the relevance attribute of 78.67%, the novelty of 90.67%, the serendipity of 75.33%, and increasing recommendation diversity of 80%.

Furthermore, the performance of the elective course recommendation system is evaluated objectively using an accuracy measurement. Comparative tests were also carried out between systems using weighted cosine similarity and those using cosine similarity. Experimental result shows that the proposed recommendation system has outperformed for both second and third-year student respondents (Fig. 8).

IV. CONCLUSION

This study proposes an elective courses recommendation system based on the value of student courses, called similarity weights. The experimental results show that the weighted recommendation system outperformed than without weights, both subjectively and objectively.

However, the proposed recommendation system can be improved by considering other factors, both in terms of students and the methods used. Several important factors can be considered to build a better recommendation system, such as class schedules, lecturers, and student research interests. Furthermore, another thing that becomes a challenge is how to handle the opportunity to combine several curricula, regarding the MBKM program.

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REFERENCES

- [1] Samrit, N.B. and Thomas, A., "A Recommendation System for Prediction of Elective Subjects," *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, vol. 5, no. 4, pp.36-43, 2017.
- [2] M. S. Sulaiman, A. A. Tamizi, M. R. Shamsudin, and A. Azmi, "Course recommendation system using fuzzy logic approach," *Indones. J. Electr. Eng. Comput. Sci.*, vol. 17, no. 1, pp. 365–371, 2019, doi: 10.11591/ijeecs.v17.i1.pp365-371.
- [3] C. C. Aggarwal, *Recommender Systems*. Cham: Springer International Publishing, 2016.
- [4] S. Asadi, S. M. Jafari, and Z. Shokrollahi, "Developing a Course Recommender by Combining Clustering and Fuzzy Association Rules," *J. AI Data Min.*, vol. 7, no. 2, pp. 249–262, 2019, doi: 10.22044/JADM.2018.6260.1739.
- [5] G. Ferio, R. Intan, and S. Rostianingsih, "Sistem Rekomendasi Mata Kuliah Pilihan Menggunakan Metode User Based Collaborative Filtering Berbasis Algoritma Adjusted Cosine Similarity," *J. Infra*, vol. 7, no. 1, pp. 1–7, 2019.
- [6] Parwita, W. G. S. and Winarko, E., "Hybrid Recommendation System Memanfaatkan Penggalan Frequent Itemset dan Perbandingan Keyword", *IJCCS (Indonesian Journal of Computing and Cybernetics Systems)*, vol. 9, no. 2, pp. 176, 2015, doi: 10.22146/ijccs.7545.
- [7] D. Worth, "Introduction to Modern Information Retrieval, 3rd Edition," *Aust. Acad. Res. Libr.*, vol. 41, no. 4, pp. 305–306, Dec. 2010, doi: 10.1080/00048623.2010.10721488.
- [8] S. Kannan *et al.*, "Preprocessing Techniques for Text Mining," *Int. J. Comput. Sci. Commun. Networks*, vol. 5, no. 1, pp. 7–16, 2015.
- [9] Uysal, A.K. and Gunal, S., The impact of preprocessing on text classification, *Information Processing & Management*, vol. 50, no. 1, pp.104-112, 2014.
- [10] J. Kanerva, F. Ginter, and T. Salakoski, "Universal Lemmatizer: A sequence-to-sequence model for lemmatizing Universal Dependencies treebanks," *Nat. Lang. Eng.*, pp. 1–30, 2020, doi: 10.1017/S1351324920000224.
- [11] F. O. Isinkaye, Y. O. Folajimi, and B. A. Ojokoh, "Recommendation systems: Principles, methods and evaluation," *Egyptian Informatics Journal*, vol. 16, no. 3. Elsevier B.V., pp. 261–273, Nov. 01, 2015, doi: 10.1016/j.eij.2015.06.005.
- [12] Munot, N. and Govilkar, S.S., 2014, Comparative study of text summarization methods, *International Journal of Computer Applications*, 102(12).